
Decodability Is Not Localization: Auditing Hidden-State Error Probes in Qwen2.5-Math-1.5B-Instruct

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 We audit whether hidden-state probes identify when mathematical reasoning first
2 becomes invalid, or instead exploit easier correlates of an invalid prefix. In a case
3 study of Qwen2.5-Math-1.5B-Instruct on ProcessBench, a linear probe predicts
4 the persistent target *invalid so far* with held-out AUROC 0.866. Equally tuned
5 nuisance baselines collapse most of that margin: prefix TF-IDF reaches 0.751,
6 structural metadata 0.776, and a joint text-plus-metadata model with final-answer
7 correctness reaches 0.874—numerically *above* the hidden state (paired difference
8 -0.008 , 95% CI $[-0.028, 0.013]$). The hidden probe’s remaining value is nar-
9 rower and task-specific: it localizes the annotated onset better (Process F1 0.393
10 versus 0.294; paired difference 0.099 $[0.040, 0.158]$), but the oracle-assisted joint
11 model has higher correct-trace rejection (0.895 versus 0.612). The hidden probe
12 localizes exactly on only 28.9% of erroneous traces and falsely alarms on 38.8%
13 of correct traces. A post-hoc, metadata-matched transition probe obtains AUROC
14 0.769 on held-out pairs; matched-covariate balance diagnostics and reuse quan-
15 tification keep it explicitly exploratory. Two verdict readouts fail their behavioral
16 validity checks, so the intervention remains a failed pilot. We conclude that de-
17 codability, incremental information over tuned nuisances, localization, calibration,
18 and demonstrated causal use are distinct validation levels, and we extract a con-
19 crete checklist for when an internal pattern is safe to treat as a candidate discovery.

20 1 Introduction

21 Process supervision asks whether each step of a solution is valid, rather than whether the final answer
22 happens to be correct (Cobbe et al., 2021; Lightman et al., 2023). A related mechanistic question is
23 whether a solving model’s own representation changes when its reasoning first goes wrong. Probe
24 accuracy alone cannot answer it: if every boundary after the first error is labeled positive, a classifier
25 may recognize accumulated inconsistency, solution length, generator style, or final-answer outcome
26 without responding at the onset.

27 The *Interpretability for Discovery* framing makes the stakes explicit: a linearly decodable pattern
28 is a *candidate*, not yet discovered knowledge. Turning a candidate into a claim requires validation
29 at distinct levels, and failing any one changes what the pattern can be used for. This paper is a
30 worked negative example of that validation sequence. We test four claims that are often collapsed:
31 (i) invalid prefixes are linearly decodable; (ii) a score localizes the first error; (iii) the signal transfers
32 across benchmark sources; and (iv) the model behaviorally uses the decoded direction. We test them
33 in one bounded setting: Qwen2.5-Math-1.5B-Instruct on ProcessBench (Yang et al., 2024; Zheng
34 et al., 2025), with problem-grouped splits and a persistent invalid-so-far target, audited by prefix-
35 text, metadata, and joint nuisance controls fit under the same validation budget as the probe, source
36 and generator stratification, an onset-aligned analysis, and a matched transition probe.

Table 1: Closest hidden-state verifier work. “Use” records the reported downstream or causal test, not a judgment of overall paper quality.

Work	Supervision / task	Scope	Onset evaluation	Use test
Sun et al.	arithmetic error	addition, GSM8K	no first-error metric	re-prompting
ReProbe	step credibility	math, planning, QA	step scoring	test-time scaling
STEP	step quality	reasoning traces	early step scoring	pruning
Ours	human first error	four ProcessBench sources	first crossing + transition	failed pilot

The audit changes the interpretation. The hidden-state probe’s pooled AUROC is high, but an equally tuned joint shortcut model combining prefix text, structural metadata, and final-answer correctness exceeds it numerically. What survives is a narrower, task-specific advantage: trace-level localization and correct rejection, both weak in absolute terms, with poor length- and source-dependent calibration and an uninterpretable behavioral assay. We contribute a documented failure mode: strong prefix decoding and cross-source ranking can coexist with no incremental pooled information over tuned nuisances.

2 Related work and contribution boundary

External process reward models learn step quality from human or model supervision (Lightman et al., 2023; Li et al., 2023). ProcessBench annotates the earliest error in 3,400 model-generated traces from GSM8K, MATH, OlympiadBench, and Omni-MATH (Zheng et al., 2025; Hendrycks et al., 2021; He et al., 2024; Gao et al., 2024). We repartition these benchmark sources for representation analysis; they are heterogeneous sources, not independent domains or evidence of model-family generalization.

Closest to this work, Sun et al. probe arithmetic errors and use the probe for selective re-prompting; ReProbe learns hidden-state step scorers for test-time scaling; and STEP uses hidden-state scores to prune reasoning traces (Sun et al., 2025; Ni et al., 2026; Liang et al., 2026). Table 1 states the comparison at the level needed for our claim: our novelty is not that hidden states predict reasoning quality, but the joint audit of a human first-error target using first-crossing, matched-transition, source-transfer, equally tuned nuisance, and failed-assay evidence.

Probing work cautions that decodability need not imply behavioral use (Hewitt and Liang, 2019; Belinkov, 2022; Elazar et al., 2021); an additive intervention is evidence only if its behavioral readout first distinguishes the relevant baseline states.

3 Experimental design

3.1 Data flow, model, and split

Each ProcessBench trace supplies a problem, generator identity, written steps, source, final-answer correctness, and a human first-error index e_i , with $e_i = -1$ for fully correct traces. Traces sharing normalized problem text form a group. Seed 42 assigns groups within source $\times$ error-status strata to 60% train, 20% validation, and 20% test: of 3,400 attempted traces, 3,360 are retained (2,187 with an annotated error), yielding 24,909 written-step boundaries and 669 held-out traces. A 2,048-token limit excludes a complete trace rather than truncating it. Appendix A gives the full attempted/retained accounting by split and source.

We run Qwen2.5-Math-1.5B-Instruct in bfloat16 and record its 1,536-dimensional residual-stream state at each written step and each of 29 hidden-state indices. The completed extraction used batch size 16 on one NVIDIA A100 GPU. Locked package versions, compute, storage, and licenses appear in Appendix D; artifact identity fixes the model, context length, dtype, and dataset SHA-256.

3.2 Persistent target, probe, and trace-level estimands

At boundary k of trace i , the frozen primary target is

$$y_{ik} = \mathbb{I}[e_i \geq 0 \wedge k \geq e_i]. \quad (1)$$

It means *invalid so far*; it does not label only the first invalid step. A class-balanced, L2 logistic probe is fit separately at each index after train-set standardization. Validation AUROC selects $C \in \{0.01, 0.1, 1, 10\}$; validation Process F1 selects a threshold on a 0.005 grid from 0.05 to 0.95 and then the layer. The selected index is 23, $C = 0.01$, threshold 0.645; it is refit on train plus validation and evaluated once on test.

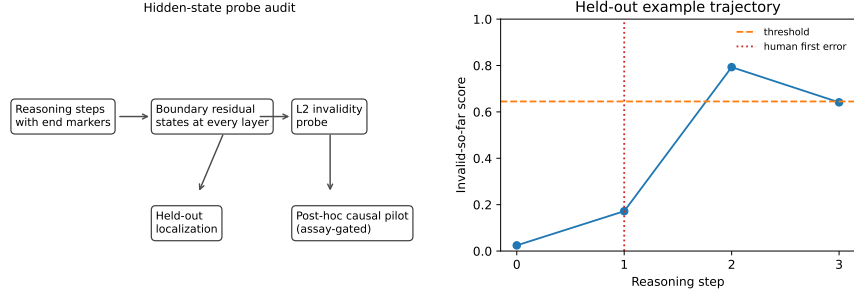


Figure 1: The persistent target and first-crossing evaluation. A causal forward pass produces one boundary state per step. In the example, the score crosses one step after the annotated error.

80 The predicted first error is the first threshold crossing, or -1 if none occurs. We report boundary
 81 AUROC and average precision, exact error-boundary accuracy among erroneous traces, correct-trace
 82 rejection, their harmonic mean (Process F1), complete-trace accuracy, and detection within one or
 83 two steps; the localization metrics weight traces equally. Primary intervals use 1,000 whole-trace
 84 bootstrap draws; the post-hoc paired comparisons use 2,000 draws because they were added in a
 85 later audit configuration, which we state rather than presenting one frozen protocol.

86 3.3 Shortcut, joint-control, and weighting audits

87 Every nuisance control receives the same model-selection budget as the hidden-state probe: L2 reg-
 88 ularization is selected from $\{0.01, 0.1, 1, 10\}$ by validation AUROC, the threshold is selected on the
 89 same validation scores, and the selected model is refit on train plus validation before the single test
 90 evaluation. Each control is also fit with sample weights inverse to boundary counts, so every trace
 91 contributes equal total training loss, mirroring a trace-equal objective rather than reweighting evalua-
 92 tion after fitting. Current-step TF-IDF uses the destination step; prefix TF-IDF uses the problem and
 93 all reasoning text through the current boundary; structural metadata uses source, generator, absolute
 94 and fractional step position, trace length, and token count. Two *joint* baselines combine prefix text,
 95 all structural metadata, and—in the stronger variant—final-answer correctness, so complementarity
 96 across separately fit controls cannot be mistaken for a hidden-state advantage.
 97 Final-answer correctness is a benchmark annotation, not an online signal: it records whether the
 98 trace’s final answer matches the benchmark reference solution, for every trace regardless of where
 99 an error occurs. We use it as a diagnostic upper bound on nuisance predictability, not a deployable
 100 baseline. A shuffled-label hidden probe checks the fitting procedure. We report paired probe-minus-
 101 control intervals from whole-trace resampling, pooled and within final-answer-correctness strata.
 102 The primary objective weights each boundary equally in *evaluation*; the trace-equal refit above
 103 addresses the training asymmetry. Descriptive subgroup outcomes stratify by source, generator,
 104 final-answer correctness, error position, and length and token-count quartiles.

105 3.4 Post-hoc onset analyses

106 The following analyses were designed after the primary test result and are labeled post-hoc. We cir-
 107 cularly shift each frozen score trajectory 5,000 times, preserving its values while breaking alignment
 108 to the annotation. We compare the score jump at each noninitial error with a transition from a fully
 109 correct trace, preferring the same source and generator and then minimizing differences in relative
 110 position, trace length, and token count.

111 The transition probe is a separate analysis of $h_{i,e_i} - h_{i,e_i-1}$. Every eligible erroneous transition is
 112 paired with a correct-trace transition in the same frozen partition, giving 1,152 train, 387 validation,
 113 and 380 test pairs; the pairing reuses correct traces (median 2 pairs per placebo trace, 90th percentile
 114 6, maximum 46, from 687 unique placebos), and matched groups are close on relative position but
 115 retain token-count and length differences (standardized differences 0.28 and 0.15). Layer and C are
 116 selected by validation AUROC; index 21 and $C = 0.01$ are selected. Controls predict from position,
 117 destination-step TF-IDF, or shuffled labels. The protocol is fully specified and executable, but the
 118 test set was not untouched when the analysis was conceived, so the result is exploratory regardless
 119 of its interval.

Table 2: Held-out comparison generated from the audited metric schema. Controls use the same validation C grid, validation threshold selection, and trace-equal training weights; the frozen hidden probe retains its original boundary-equal fit. Paired intervals are hidden minus control; positive Δ favors the hidden state.

Predictor	AUROC	Proc. F1	Error exact	Corr. rej.	Δ AUROC [95% CI]
Hidden state	.866	.393	.289	.612	—
Prefix TF-IDF	.751	.274	.192	.477	.115 [.089,.142]
Structural metadata	.776	.168	.148	.194	.090 [.069,.110]
Joint text + metadata	.810	.210	.169	.278	.056 [.037,.073]
Metadata + final outcome	.854	.262	.164	.646	.012 [−.012,.036]
Joint + final outcome	.874	.294	.176	.895	−.008 [−.028,.013]

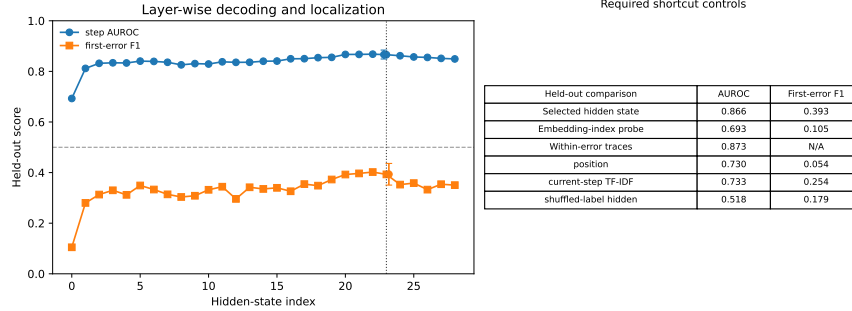


Figure 2: Layer-wise held-out performance and original controls. The embedding-index row is a linear probe at hidden-state index 0, included as a depth comparison rather than a lexical control. Process F1 is N/A within error-only traces because correct-trace rejection is undefined. The equally tuned controls appear in Table 2.

4 Results

4.1 A tuned joint shortcut model matches or exceeds pooled AUROC

The selected probe reaches AUROC 0.866 (95% CI [0.849, 0.884]), average precision 0.843 ([0.813, 0.870]), and step F1 0.743 ([0.716, 0.769]). Within erroneous traces, first-step centering leaves pooled AUROC at 0.881 ([0.862, 0.900]), and mean per-trace AUROC is 0.968 ([0.958, 0.976]). Stable trace offsets therefore do not explain all discrimination.

Table 2 shows the equally tuned comparison. Prefix text reaches AUROC 0.751, structural metadata 0.776, and their joint model 0.810: the controls’ errors are partly complementary. Adding final-answer correctness yields the strongest nuisance model at AUROC 0.874, numerically above the hidden probe’s 0.866 (paired hidden-minus-joint difference -0.008 [−0.028, 0.013], which includes zero). Under a maximally fair tuning budget, the hidden hidden-only comparison does not establish whether hidden features add incremental value to the nuisance model; that requires the nested N versus N+H comparison described in the limitations.

The decomposition matters more than the pooled number. The hidden probe retains a Process F1 advantage of 0.099 ([0.040, 0.158]) and an erroneous-trace exact-localization advantage of 0.113 ([0.064, 0.163]). This gain is offset by lower correct rejection: 0.612 for the hidden probe versus 0.895 for the oracle-assisted joint model (hidden-minus-joint difference -0.283 [−0.347, −0.218]). On the 237 correct traces, both reject 140, only the hidden probe alarms on 72, only the joint model alarms on 5, and both alarm on 20. The models therefore trade erroneous-trace onset localization against correct-trace rejection. Within final-answer-correctness strata, the hidden AUROC advantage is confined to traces whose final answer is correct (0.073 [0.039, 0.107]) and vanishes among incorrect-outcome traces (-0.001 [−0.022, 0.021]), consistent with the outcome field carrying most pooled information when it is informative.

Equal trace weighting at evaluation changes AUROC by +0.001 (95% paired CI [−0.007, 0.010]). Complete-trace accuracy falls from 0.491 in the shortest length quartile to 0.264 in the longest, with heterogeneous source-level estimates (Appendix B).

4.2 First-error localization remains weak

The hidden score localizes 28.9% of erroneous traces exactly, 59.3% within one step, and 72.7% within two. It crosses somewhere on 88.7% of erroneous traces, but also on 38.8% of fully correct traces, and among detections it is late by +0.59 steps on average ([+0.36, +0.80]). Process F1 is 0.393 ([0.351, 0.436]); complete-trace accuracy is 0.404.

Post-hoc temporal tests show more alignment than these threshold metrics alone suggest. Exact localization is 0.289 versus 0.164 under circular shifts ($p = 0.0002$). Across 380 noninitial errors, the mean onset jump is 0.257; matched correct transitions rise 0.113, for a paired difference of 0.144 ([0.096, 0.193]). The score continues from 0.341 immediately before the error to 0.559 at onset, 0.680 one step later, and 0.731 two steps later: a gradual transition into a persistent state, not a sharp alarm. The transition task excludes 52 test traces whose first error occurs at step 0, which may differ systematically from the 380 eligible ones.

The matched transition probe reaches AUROC 0.769 ([0.713, 0.820]), average precision 0.741 ([0.672, 0.816]), and paired accuracy 0.753 ([0.676, 0.824]); position, destination-step TF-IDF, and shuffled-label controls reach 0.63, 0.67, and 0.58. Given the placebo reuse and retained covariate imbalance documented above, this supports decodability of an onset-related change on the already-inspected split; sensitivity refits under one-to-one and inverse-reuse matching, and replication on untouched data, are needed before any confirmatory reading.

4.3 Source ranking transfers; model generalization is untested

Separate probes trained on one ProcessBench source and tested on another have mean off-diagonal AUROC 0.805 versus 0.845 on the diagonal; every off-diagonal cell exceeds 0.739. Thresholded Process F1 is less stable: 0.261 off diagonal versus 0.336 on diagonal, with individual off-diagonal values from 0.055 to 0.371. These cells show cross-source ranking within one benchmark construction, not transfer to another model size, family, or dataset.

4.4 Scope of the evidence

This paper is an existence proof of a failure mode in one small math-tuned model on one benchmark construction—decodability without incremental pooled information, localization, calibration, or causal validation—not a measurement of its prevalence across models. A second-model replication under the frozen protocols reported here is deliberately left as follow-up work.

5 Limitations, responsible use, and conclusion

This is a one-model, one-benchmark case study; the title and claims name Qwen2.5-Math-1.5B-Instruct. All added analyses—transition probe, temporal randomization, shortcut and joint controls, trace-equal refits, subgroups—are post-hoc, and all intervals are conditional on one frozen split and the completed selection pipeline: they capture resampling uncertainty, not split or selection sensitivity. No dated preregistration exists. Corrected/error twins lack independent verification, so no patching result is reported; a stronger follow-up would verify twins, reserve an untouched onset set, and validate the readout first.

The detector is unsuitable for grading, certification, or unattended feedback: its 38.8% false-alarm rate on correct traces could penalize valid work, its late detections could let an invalid derivation propagate, and its sensitivity to source, outcome, and length makes distribution shift especially risky. Positive uses are limited to research diagnostics with human or formal verification.

The central empirical finding is a gap with a precise shape: in Qwen2.5-Math-1.5B-Instruct, invalid-so-far prefixes are linearly decodable, but an equally tuned joint text-plus-metadatum model with final-answer correctness matches or exceeds the hidden state’s pooled AUROC; what remains is a better erroneous-trace onset-localization score but worse correct rejection, both weak in absolute terms. A nested nuisance-plus-hidden analysis is required before assigning that contrast uniquely to the hidden representation.

For discovery practice, this audit reduces to a checklist. Before treating a decodable internal pattern as discovered knowledge, test and report: (1) whether adding hidden features to equally tuned nuisance models—under the same selection budget—improves held-out performance; (2) whether any residual advantage survives on the task-specific metrics that matter, with paired intervals; (3) whether temporal correspondence holds against matched and shifted controls; (4) whether calibration and subgroup behavior survive the shifts the use case implies; and (5) whether a behavioral readout distinguishes the relevant states before any causal claim. Our candidate has not yet com-

pleted the nested test in (1), shows metric-specific rather than uniformly strong performance in (2)–(4), and fails the behavioral-readout requirement in (5).

References

- Yonatan Belinkov. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219, 2022. doi: 10.1162/coli_a_00422.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021. URL <https://arxiv.org/abs/2110.14168>.
- Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *Transactions of the Association for Computational Linguistics*, 9:160–175, 2021. doi: 10.1162/tac1_a_00359. URL <https://aclanthology.org/2021.tac1-1.10/>.
- Bofei Gao, Feifan Song, Zhe Yang, Zefan Cai, Yibo Miao, Qingxiu Dong, Lei Li, Chenghao Ma, Liang Chen, Runxin Xu, Zhengyang Tang, Benyou Wang, Daoguang Zan, Shanghaoran Quan, Ge Zhang, Lei Sha, Yichang Zhang, Xuancheng Ren, Tianyu Liu, and Baobao Chang. Omni-MATH: A universal olympiad-level mathematics benchmark for large language models. *arXiv preprint arXiv:2410.07985*, 2024. URL <https://arxiv.org/abs/2410.07985>.
- Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-Bench: A challenging benchmark for promoting AGI with olympiad-level bilingual multimodal scientific problems. *arXiv preprint arXiv:2402.14008*, 2024. URL <https://arxiv.org/abs/2402.14008>.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. *arXiv preprint arXiv:2103.03874*, 2021. URL <https://arxiv.org/abs/2103.03874>.
- John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, pages 2733–2743. Association for Computational Linguistics, 2019. doi: 10.18653/v1/D19-1275. URL <https://aclanthology.org/D19-1275/>.
- Yifei Li, Zeqi Lin, Shizhuo Zhang, Qiang Fu, Bei Chen, Jian-Guang Lou, and Weizhu Chen. Making language models better reasoners with step-aware verifier. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, pages 5315–5333. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.291. URL <https://aclanthology.org/2023.acl-long.291/>.
- Zhixiang Liang, Beichen Huang, Zheng Wang, and Minjia Zhang. Hidden states as early signals: Step-level trace evaluation and pruning for efficient test-time scaling. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 26800–26813. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1336. URL <https://aclanthology.org/2026.findings-acl.1336/>.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. *arXiv preprint arXiv:2305.20050*, 2023. URL <https://arxiv.org/abs/2305.20050>.
- Jingwei Ni, Ekaterina Fadeeva, Tianyi Wu, Mubashara Akhtar, Jiaheng Zhang, Elliott Ash, Markus Leippold, Timothy Baldwin, See-Kiong Ng, Artem Shelmanov, and Mrinmaya Sachan. ReProbe: Efficient test-time scaling of multi-step reasoning by probing internal states of large language models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 11667–11689. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.536. URL <https://aclanthology.org/2026.acl-long.536/>.
- Yucheng Sun, Alessandro Stolfo, and Mrinmaya Sachan. Probing for arithmetic errors in language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 8111–8128. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.411. URL <https://aclanthology.org/2025.emnlp-main.411/>.

254 An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu,
 255 Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu,
 256 Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math technical report: Toward mathematical
 257 expert model via self-improvement. *arXiv preprint arXiv:2409.12122*, 2024. URL <https://arxiv.org/abs/2409.12122>.
 258
 259 Chujie Zheng, Zhenru Zhang, Beichen Zhang, Runji Lin, Keming Lu, Bowen Yu, Dayiheng
 260 Liu, Jingren Zhou, and Junyang Lin. ProcessBench: Identifying process errors in mathemat-
 261 ical reasoning. In *Proceedings of the 63rd Annual Meeting of the Association for Compu-*
 262 *tational Linguistics*, pages 987–1011. Association for Computational Linguistics, 2025. doi:
 263 10.18653/v1/2025.acl-long.50. URL <https://aclanthology.org/2025.acl-long.50/>.

264 A Additional data accounting

Table 3: Data flow. Attempted / retained traces / exclusions by source and split; boundary counts are written-step boundaries.

Split	Attempted	Retained	Excluded	Boundaries	Error traces	Correct traces
Train	2,041	2,014	27	14,944	1,311	703
Validation	681	677	4	4,980	444	233
Test	678	669	9	4,985	432	237
All	3,400	3,360	40	24,909	2,187	1,173

Table 4: Attempted / retained traces / exclusions by source and split.

Split	GSM8K	MATH	OlympiadBench	Omni-MATH
Train	236/236/0	597/592/5	602/594/8	606/592/14
Validation	80/80/0	198/198/0	204/202/2	199/197/2
Test	84/84/0	205/204/1	194/188/6	195/193/2

265 The retained boundary counts by split and source are, respectively: train 1,229 / 3,718 / 5,129 /
 266 4,868; validation 414 / 1,303 / 1,708 / 1,555; and test 439 / 1,387 / 1,661 / 1,498 in the source
 267 order of the table. The original attempted total was 25,697 boundaries; 24,909 remain after the 40
 268 complete-trace exclusions.

269 B Descriptive subgroup audit

Table 5: Trace-macro outcomes by ProcessBench source (95% whole-trace bootstrap intervals).

Source	Complete accuracy	Exact error	Correct rejection
GSM8K	.512 [.405,.619]	.238 [.116,.375]	.786 [.658,.902]
MATH	.500 [.426,.569]	.347 [.264,.435]	.723 [.620,.818]
OlympiadBench	.298 [.234,.362]	.207 [.138,.282]	.463 [.340,.585]
Omni-MATH	.358 [.290,.425]	.324 [.253,.401]	.467 [.319,.614]

270 Among traces with an incorrect final answer, complete accuracy is 0.292 ([0.245, 0.342]). Among
 271 traces with a correct final answer it is 0.506 ([0.451, 0.554]); this group mixes valid traces and traces
 272 whose reasoning contains an annotated error. Across generators with at least 20 held-out traces,
 273 complete accuracy ranges from 0.316 to 0.621. These wide, overlapping estimates are descriptive
 274 evidence of heterogeneity rather than generator rankings.

275 C The failed behavioral pilot

276 The original CORRECT/INCORRECT readout has baseline AUROC 0.283 and specificity zero. A re-
 277 vised question with conditional Yes/No probabilities remains below chance (AUROC 0.342, speci-
 278 ficity 0.203 on 256 balanced boundaries). The intervention code requires both AUROC > 0.5 and

nonzero specificity before a learned-direction effect is interpreted. Neither readout passes. Random orthogonal directions cannot substitute for a positive control of assay sensitivity, so the dose response is not causal evidence for or against behavioral use. We report the pilot only to delimit the evidence: no directional intervention was interpreted, so the causal section of the audit is empty by design rather than by null result.

D Reproduction record

The dataset SHA-256 is 447f0a4b35c5747a9f9a3dab1e70d43f71efd501497b8cec668b71337099784a. All stochastic analysis uses seed 42. Exact resolved YAML files and artifact identity files are included with the anonymous supplement. The principal commands are:

```
uv sync --extra dev
uv run math-error-tracing \
  --config configs/experiment1.yaml validate-config
uv run math-error-tracing \
  --config configs/experiment1.yaml run-all
uv run math-error-cpu-followup \
  --config configs/experiment2.yaml
uv run math-error-extended-followup \
  --config configs/experiment3.yaml \
  fit-transition-probe
uv run math-error-extended-followup \
  --config configs/experiment3.yaml \
  analyze-transition-matching
uv run pytest
uv run ruff check src tests
```

Table 6: Compute and licensing record.

Extraction and interventions	one NVIDIA A100 (40 GB), bfloat16, batch 16; single passes over 3,400 traces; ~hours of GPU time
Probe fitting and audits	CPU only; the largest stage bootstraps 2,000 trace resamples
Storage	~1 GB compact predictions plus shards for raw activations
Locked versions	Python 3.10+; PyTorch 2.13.0; Transformers 4.57.6; scikit-learn 1.7.2; NumPy 2.2–2.5; pandas 2.3.3; SciPy 1.15–1.18; datasets 4.8.5
Licenses	ProcessBench: Apache 2.0. Qwen2.5-Math-1.5B-Instruct: Apache 2.0. Analysis code: repository license.
	Benchmark sources retain their original terms.

Activation extraction and the original intervention used one A100 GPU; probe fitting and all audit analyses are CPU-capable. Extraction is sharded and resumable. The code rejects mismatched dataset, model, dtype, and context-length identities and stores all selected thresholds and predictions. The supplement records the exact environment of each analysis stage in its machine-readable summary files.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Introduction name the single model and benchmark, distinguish the persistent prefix target from onset detection, and state the negative and post-hoc results.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: The final main-text section discusses one-model scope, shortcut sensitivity, post-hoc analyses, the absence of preregistration, and the invalid causal assay.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate “Limitations” section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren’t acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper contains no theoretical results or proof claims.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 and Appendix D specify the split, target, selection protocol, seeds, locked package versions, dataset identity, and exact reproduction commands. The joint shortcut controls share the hidden probe's validation budget, and the transition-matching protocol is stated with its reuse diagnostics.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
 - (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: An anonymized code supplement includes the pipelines, tests, configurations, and instructions; the upstream dataset and model are publicly accessible.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 reports the data partitions, exclusions, model and activation location, probe family, hyperparameter grid, threshold selection, controls, and estimands.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: The main results use 95% whole-trace bootstrap intervals, paired control differences, and a two-way clustered bootstrap for reused transition controls.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix D, Table 6 records the GPU model, precision, batch size, approximate per-stage compute, and storage footprint. Exact wall-clock logs for the original extraction were not preserved; the paper states the order of magnitude instead of inventing precision.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work analyzes publicly released model and benchmark artifacts, makes bounded claims, preserves submission anonymity, and reports failures and limitations.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. Broader impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: The responsible-use discussion identifies research-diagnostic value and the concrete risks of false alarms, late detections, and distribution shift in grading uses.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The paper releases no new pretrained model, scraped dataset, or other asset with a high misuse risk; it releases analysis code for public research artifacts.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: Appendix D, Table 6 records the Apache 2.0 licenses of the ProcessBench dataset and the Qwen model, and notes that the benchmark’s constituent sources retain their original terms; the paper also cites both upstream artifacts.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.

- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [\[Yes\]](#)

Justification: The anonymous supplement documents the analysis code, resolved configurations, artifact schemas, commands, tests, and known limitations.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [\[N/A\]](#)

Justification: This study performs no crowdsourcing and recruits no human subjects.

Guidelines:

- The answer [\[N/A\]](#) means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. Institutional review board (IRB) approvals or equivalent for research with human subjects

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [\[N/A\]](#)

Justification: This study performs no human-subject research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. Declaration of LLM usage

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: The analyzed language model is the experimental subject, not a non-standard agent used to generate labels, hypotheses, or core methodological decisions. Agent assistance contributed to code, analysis, and manuscript editing under author direction; the authors verified all reported numbers against repository artifacts and remain responsible for correctness and originality.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.